

Design a Rate Limiter

1. Introduction

In a network system, a **rate limiter** is used to control the rate of traffic sent by a client or a service.

In the HTTP world, a rate limiter limits the number of client requests allowed to be sent over a specified period.

None

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If the API request count exceeds the defined threshold,  
all excess calls are blocked.
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2. Example Use Cases

None

- A user can write no more than 2 posts per second.
- You can create a maximum of 10 accounts per day from the same IP address.
- You can claim rewards no more than 5 times per week from the same device.

3. Benefits of Rate Limiter

✓ Prevent Resource Starvation (DoS Protection)

A rate limiter helps prevent resource starvation caused by Denial of Service (DoS) attacks.

Almost all APIs published by large tech companies enforce some form of rate limiting.

None

- Twitter limits the number of tweets to 300 per 3 hours.
- Google Docs APIs: 300 requests per user per 60 seconds (read requests).

A rate limiter blocks excess calls, whether they are:

None

- Intentional (malicious attacks)
- Unintentional (traffic spikes)

✓ Reduce Cost

Limiting excess requests reduces the number of required servers and allows allocating resources to high-priority APIs.

Rate limiting is especially important when using paid third-party APIs.

None

Examples:

- Check credit
- Make a payment
- Retrieve health records

These APIs are charged on a per-call basis, so limiting calls helps reduce costs.

✓ Prevent Server Overload

To reduce server load, a rate limiter filters out excess requests caused by:

None

- Bots
- User misbehavior

This helps maintain system stability.

4. Summary

A rate limiter:

None

- Controls request rate
- Blocks excess traffic
- Protects system resources
- Reduces operational cost
- Prevents server overload